Introduction and Math Review Practice 3

Module 1 - Introduction & Math Review · Daily practice sheet · No date

Name _____ Date _____

1a 1. The table below shows the decomposition of a drug as a function of time.

Time (minute)	Drug A (mg)
10	97.0
20	89.0
40	73.0
60	57.0
90	34.0
120	10.0
130	2.5

a. How would you classify the decrease in the amount of drug A?

A. First-order
B. Neither order fits the data
C. Zero-order

1b 1. The table below shows the decomposition of a drug as a function of time.

Time (minute)	Drug A (mg)
10	97.0
20	89.0
40	73.0
60	57.0
90	34.0
120	10.0
130	2.5

b. What is the rate constant for the decrease in drug?

1c 1. The table below shows the decomposition of a drug as a function of time.

Time (minute)	Drug A (mg)
10	97.0
20	89.0
40	73.0
60	57.0
90	34.0
120	10.0
130	2.5

c. What is the half-life of drug decomposition?

2a 2. A solution of a drug was freshly prepared at a concentration of 500 mg/mL. After 60 days at 25°C, the drug concentration in the solution was 150 mg/mL. a. Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

2b 2. A solution of a drug was freshly prepared at a concentration of 500 mg/mL. After 60 days at 25°C, the drug concentration in the solution was 150 mg/mL. b. Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?